Radu Andrei Tudorica

Bucharest, Romania | raduandreitudorica3@gmail.com | 0799 159 414 | linkedin.com/in/raduandreitudorica |
github.com/RaduAndreiTudorica

Education

National University of Science and Technology POLITEHNICA Bucharest — Sept 2024 – Present
Bachelor of Science in Computer Science (ACS CTI) — Bucharest, Romania

- **Relevant Coursework:** Algorithm Analysis, Data Structures, Computer Programming (C), Operating Systems, Assembly Language & Computer Architecture, Computer Networks, Communication Protocols, Numerical Methods

Experience

Google Summer of Code 2026 – Applicant — Apr 2026
Open-Source Contributor — Remote

- Submitted a project proposal for GSoC 2026; results pending (April 30, 2026).
- Studied the target organization's codebase, contributed to discussions, and prepared a detailed implementation plan.

Projects

ACSQL – In-Memory Database & Query Engine | *C, Memory Management, Cryptography* | GitHub May 2024 – Jun 2024

- Developed an in-memory relational database management system in C, supporting complex entity relationships and dynamic schema management.
- Implemented a custom query engine with support for filtered searches, conditional updates, and cascading deletes across related tables.
- Engineered a **Cipher Block Chaining (CBC)** encryption module from scratch using bitwise operations and custom S/P-box substitution tables.
- Achieved zero memory leaks across all operations – verified via **Valgrind** with rigorous tracking of nested dynamically allocated structures.

Minecraft Chunk Engine (libchunk) | *C, Advanced Pointers, Bitwise Optimization* | GitHub Nov 2024 – Dec 2024

- Built a C library for 3D spatial data manipulation, representing Minecraft chunks via multi-dimensional pointer structures (`char***`).
- Implemented a custom **Run-Length Encoding (RLE)** compression algorithm using advanced bitwise shifts and masking to reduce memory footprint.
- Engineered recursive matrix transformations for gravity simulation and procedural generation of 3D geometries (spheres, cuboids).
- Managed complex dynamic memory reallocation routines with zero leaks across all execution paths.

Lox Interpreter – Java Implementation | *Java, Compiler Architecture, AST, Parsing* | GitHub Jul 2025 – Present

- Built a complete interpreter from scratch: lexical scanner, recursive descent parser, and tree-walk evaluator.
- Implemented runtime support for variable scoping, first-class functions, closures, and control flow.
- Applied the **Visitor Pattern** for clean separation between AST structure and evaluation logic.

Technical Skills

Languages: C, C++, Bash, Java, Scala, Python, x86 Assembly, MATLAB
Build & Debug Tools: GCC, Makefile, Valgrind, GDB, Maven
Developer Tools: Git, GitHub, VS Code, Linux (Ubuntu, Arch)
Concepts: Memory Management, Systems Programming, Cryptography, Compiler Design, Data Structures & Algorithms
Scripting & Automation: Bash scripting, CLI tool development